



COS60004 Creating Web Applications

Assignment Part 1-20% of final mark Develop a Simple Static Website

Due Date	10 am, Monday Week 6 (Late submission penalty 10% of total available marks per 24 hours)
Demonstration	on your tutorial, Before or at Week 5 (Demonstration is mandatory, penalty for assignments without demonstration is 30% of total available marks)

Note: This is an individual assignment. Generative AI may be used for learning or generating ideas but must be properly cited, including screenshots of prompts used. Submissions are subject to plagiarism checks. Unacknowledged similarities or failure to explain your work may result in penalties for academic misconduct.

Purpose

Develop a website to promote a specific product/service (assigned by tutor) for a fictional company, enabling customers to view details and make inquiries.

Essential Requirements

Required Files (exact names required):

- HTML Pages: index.html, product.html, enquire.html, about.html, enhancements.html
- CSS File: style.css (or modular CSS files starting with "style")
- Wireframe: PDF or image file showing layout for all pages

Each student will be allocated a different type of product or service by their tutor.

Notes:

- Pages must validate against W3C HTML5 and CSS3 validators.
- No deprecated elements (e.g., <i>,) or iframes.
- Use semantic HTML5 elements (e.g., <section> over <div> where appropriate).
- Consistent layout and navigation across all pages using a common menu.
- Your code must be deployed on the Mercury server with a relative path (no absolute path). The link to your Mercury website must be submitted as a comment in Canvas.
- Follow in-house templates for metadata, tables, and accessibility.
- Acknowledge all third-party material with hyperlinked references. Attach the prompts screenshots to your submission.

No JavaScript is to be used in this part of the assignment – we will use JS in Part 2.

Page-Specific Requirements

1. Introductory home page (index.html)

- Content: Company title, description, and a relevant graphic.
- Features: Background graphic, footer with small, centred text.
- Menu: Links to all pages, consistent across the site (The same menu should appear on every page of your website).

2. Product range page (product.html)

Content: Describe at least 3 products/services (real or imaginary, ≥150 words) with optional features. For example, a restaurant meal could consist of a range of courses, with dine-in or take-away options etc. A camera might come in several models, with options for colours, lenses, etc.

HTML page should contain: Hierarchical headings (≥2 levels), multiple <section> elements, <aside> with relevant content, ordered and unordered lists, and at least one image (<100KB).

CSS: <h1> styled with shorthand font property, <aside> at 25% width floating right with coloured border, margin, and padding; footer spans full page width, small centred text.

Sources / References: In this assignment, you may use material from other websites, but the source of all material must be acknowledged. This acknowledgement should be immediately after the material and include a hyperlinked URL to the original source. The text of the hyperlink reference can be a short name, but the hyperlink must work. (Normally permission must be given before re-publishing content, but our websites are not live.). *It is permitted to use AI to generate images or product descriptions.*

3. Product enquiry page (enquire.html)

This page has a form where a user can lodge an enquiry about a particular product. The form will have the following form controls and HTML5 data validation should be used to check the user's input.

Field	Format Requirement
First name	<i>type text, maximum of 25 characters, alphabetical only</i>
Last name	<i>type text, maximum of 25 characters, alphabetical only</i>
Email address	<i>validate format</i>
Address	☒ Street address: <i>type text, a maximum of 40 characters, accepting numbers, alphabets, and spaces</i> ☒ Suburb/town: <i>type text, a maximum of 20 characters, accepting numbers, alphabets, and spaces</i> ☒ State: <i>type select list (drop-down menu) with options VIC, NSW, QLD, NT, WA, SA, TAS, ACT</i> ☒ Postcode: <i>type exactly 4 digits</i> ☒ All address elements are grouped using a fieldset and legend
Phone number:	<i>type text, maximum of 10 digits. With placeholder</i>
Preferred contact	<i>(email, post, phone). use radio buttons.</i>
Product	<i>from the range the user wants to enquire about. use select list</i>
Product Features	<i>use checkboxes</i>
Comment field	<i>for example, allowing the reader to specify a particular aspect they are interested in, use textarea, use a placeholder</i>

All inputs should have labels. All form values, except the comment textarea, are 'required' or have a default value (e.g. select, checkbox, radio). **The user should not be able to submit the form** if any of these required fields are blank. Use HTML5 validation to check the format.

Data Submission to Server

This form must have a Submit input. When the submit button is pressed, the name values from the form will be sent to the server using the post HTTP method. The server action address <https://mercury.swin.edu.au/it000000/formtest.php>. The server will then just echo back the namevalue pairs to the client. While nothing will be stored on the server in this part of the assignment, (we will do this in Part 3) this will allow the form to be tested.

3. A page about you (about.html)

Content: Personal profile (e.g., resume, interests). It could also include a resume, interests, or information that is related to you. This extra information allows you to extend the techniques you apply in your assignment, and could include:

- Demographic information about you
- Description of hometown
- A list of your favourite books, music, films, etc.

The page should contain:

- Student details in a definition list (<dl>): Name, student number, tutor's name, course, and mailto link to student email.
- Photo (<100KB) in <figure>, floated right, double border, 20px padding.
- Swinburne timetable in a centred, HTML table with bold headings and hex-coloured cell backgrounds.

5. Wireframe Design Requirement

Before you begin coding your webpages, you are required to create a wireframe that visually represents the layout and structure of your website. The wireframe should help you plan the placement of key elements such as navigation, images, text content, forms, and footers across all pages.

A single wireframe or multiple wireframes (one for each page if needed), showing the structure of:

- Home page (index.html)
- Product page (product.html)
- Enquiry form page (enquire.html)
- About page (about.html)
- Enhancements page (enhancements.html)

Wireframe Format:

- You can draw your wireframe digitally using tools such as Figma, Balsamiq, Adobe XD, or even PowerPoint.
- Alternatively, you may draw it by hand, but it must be clear, labelled, and scanned or photographed neatly for submission.

Save and submit your wireframe(s) as PDF or image files in your assignment folder and upload this to Canvas along with your website code.

Wireframe Guidelines:

Include navigation/menu bar, content sections, and footers.

Clearly label page elements (e.g., header, main image, section title, enquiry form fields, etc.).

Your wireframe does not need to include styling details (colors, fonts, etc.) — it should focus on structure and layout.

6. CSS Requirements

No style markup should be included in your HTML file.

The pages in your website must be styled with CSS and have a consistent ‘look and feel’, particularly with markup of menus, headers and footers. While the emphasis in this assignment is on the appropriate application of techniques rather than graphic design, your pages should follow basic usability/accessibility principles, e.g. distinguishable foreground and background colours, font readability, etc.

You must create your own external stylesheet to implement your design. Either you should write one stylesheet for the entire website or you have to have a modular design approach based on functionality (for example, one for fonts and colours and one for the structure). It is not acceptable to have one CSS for each page.

Essentials:

- External stylesheet(s) only (style.css or modular files starting with "style").
- No inline CSS or proprietary markup (e.g., -moz-, -webkit-).
- Consistent look and feel across pages (menu, headers, footers).
- Accessibility: Readable fonts, distinguishable colours.
- All Pages: Fluid layout (reflows on resize).
- Comments: Header comments (author, date, purpose) and inline comments for clarity.
- Validate against CSS3 at <https://jigsaw.w3.org/css-validator/>.
- Selectors: Use element, #id, .class, grouping, contextual, and pseudo-class/element.
- Specific Styling:
 - Menu: Background colour.
 - Home Page: Background graphic, small centred footer text.
 - Product Page: <h1> shorthand font, <aside> styling, full-width footer.
 - About Page: Styled <dl> (aligned <dt>/<dd>), photo with double border, centred table with hex-coloured cells.

Enhancements to the Specified Requirements

Note: Complete all basic requirements before attempting enhancements.

This part of the assignment allows you to showcase your ability to research and apply advanced HTML/CSS techniques beyond lecture and tutorial content. Implement up to two enhancements within index.html, product.html, enquire.html, or about.html to improve the website’s functionality or user experience in a relevant way. On enhancements.html, list and describe each enhancement, explaining:

- How it exceeds basic requirements.
- The code used to implement it.
- Any third-party sources (e.g., URL), which must be cited.
- A hyperlink to the implementation (for CSS, link to the affected HTML).

Requirements:

Enhancements must work in Firefox.

Maximum of two enhancements assessed (up to 10 marks each).

Marks depend on novelty, completeness, and technical depth, as judged by your tutor. Trivial enhancements (e.g., a few lines of code) or common ideas used by many students score lower. Enhancements must be relevant, innovative, non-trivial, well-described, and distinct from other features.

Examples: Advanced HTML elements (e.g., image maps, <canvas>) that enhance user experience.

CSS3 features (e.g., animations, pseudo-elements like :target, child/sibling combinators, or attribute selectors) not covered in tutorials.

Web Site Folder Structure and Deployment Requirements

Create a website structured as described below. You can create additional HTML files for your content (depending on what your content requires), but the following is needed:

Use relative links only.

<code>assign1/</code>	<i>You must have this folder – case sensitive!</i>
<code>index.html</code>	
<code>product.html</code>	
<code>enquire.html</code>	
<code>about.html</code>	
<code>enhancements.html</code>	
<code>...other html pages</code>	
<code>images/</code>	<i>Folder for images for your page content</i>
<code>styles/</code>	<i>Folder for style.css and other css files</i>
<code>styles/images/</code>	<i>Folder for images referred to by your css files e.g. background</i>
<code>wireframe/</code>	<i>Folder for all the wireframe document(s)</i>

Notes:

- HTML files should only be in the base “assign1/” folder – not anywhere else.
- **All** images used for the **content** should be stored in the “assign1/images/” folder.

Assignment Submission (Mercury + Canvas)

Mercury: Deploy website, submit URL as a Canvas comment before the deadline.

Canvas: Zip all files (assign1.zip) including HTML, CSS, images, and wireframe(s). Ensure index.html runs without moving files.

- Multiple submissions allowed; only the last is marked.
- No large media files (>5MB).
- Validate HTML (<https://validator.w3.org/>) and CSS (<https://jigsaw.w3.org/css-validator/>).

Make sure you complete your Canvas submission process and thoroughly test your website deployed on the Mercury server.

Assignment 1 Mark Sheet – Assessed by tutor in tutorial

Student ID: _____ Student Name: _____ Assessor: _____ Total Marks: _____

Essential Requirements	Comment	Mark
index.html (2 marks each tick) HTML: (<i>HTML5 validation errors - deduct 1 marks each, -6 max</i>) [] - Meta info for all pages (e.g., keywords, description, charset, title) ☐ - Header and Nav that links all pages (consistent menu on all pages) ☐ CSS: Menu appropriately formatted, with background colour ☐ - Background graphic on page ☐ Footer text small and centred ☐		/10
product.html (2 marks each tick, unless specified) HTML: (<i>HTML5 validation errors - deduct 1 marks each, -8 max</i>) [] - Appropriate Headings (>2 contiguous levels) ☐ , 3+ Sections ☐ , - Ordered & Unordered list ☐ , At least three products/services ☐ CSS: <h1> font variant, size, family set, short-hand ☐ - Aside 25% width ☐ , floats right, margin, padding, coloured border ☐ - Footer full page width ☐ (1 mark)		/15
enquire.html (1 mark each tick, unless specified) HTML: (<i>HTML5 validation errors - deduct 1 marks each, -12 max</i>) [] - Form elements with HTML5 validation, with <i>required</i> or default value: Firstname & Lastname (max25 alpha) ☐ , email ☐ , street (max40 alphanumeric/space) ☐ , suburb (max20 alphanumeric/space) ☐ , state (select) ☐ , postcode (4digits) ☐ , preferred contact (radio) ☐ , Fieldset used ☐ phone (10 digits)+placeholder ☐ , product (select) ☐ , features (checkboxes) ☐ , textarea & placeholder ☐ - Labels linked with 'for' (test by 'clicking' on the label) ☐ (4 marks) - Data for all inputs returned from the server correctly ☐ (4 marks)		/20
about.html (1 mark per tick) HTML: (<i>HTML5 validation errors - deduct 1 mark each, -3 max</i>) [] - Student details via dl ☐ , figure tag, figure caption tag, photo tag ☐ , CSS: Styled dl, dt, dd ☐ photo double border, floats right, padding within figure ☐ , Table centred, table cell background hex ☐		/5
Wireframe (5 marks per tick) Structure & Layout consistent with final product (5 pages) ☐ , Sufficient labelling of elements (5 pages) ☐ , Coverage of required pages (5 pages) ☐		/15
CSS (general): (<i>CSS3 validation errors - deduct 1 mark each, -8 max</i>) [] - Selectors: element ☐ , #id ☐ , .class ☐ , contextual ☐ , pseudo ☐ , (1 mark each) - Fluid page flow when resized (uses relative dimensions) ☐ (5 marks) - All pages have a consistent and professional presentation ☐ (5 marks)		/15
Subtotal		/80

Enhancements Maximum of 2 Enhancements (the *first 2 if you have more than 2 enhancements*) will be assessed. Up to 10 marks are available per feature. **Incomplete implemented, trivial design, or vague explanation during demonstration may receive less or 0 marks.**

Feature Name	Novelty, Completeness, and Comprehensiveness	Described	Linked to implementation	Mark
		Y/N	Y/N	/10
		Y/N	Y/N	/10
Total Additions				/20

Other Deductions

Items	Max Deduction Per Page if requirements not met	Deduct
Page design (deduct up to 4 marks each) - Poorly designed structure - Inappropriate contrast in colours - Inappropriate use of fonts - Inconsistent application of style across pages - Inappropriate application of styles (e.g. different styles for menu on each page)	-20	
Content (deduct up to 3 marks each) - Product content of insufficient quantity (<150 words) - Product content poor quality - Images inappropriate, or inappropriate file sizes	-9	
HTML (deduct up to 3 marks each) - Deprecated elements/attributes have been used - Inappropriate use of HTML semantics (e.g. use of <div> when <section> <article> should be used) - HTML usability does not follow standards (e.g. alt on images, label in forms, tables) - HTML Image height and width attributes are missing or incorrect - HTML has embedded Style markup. CSS is not fully separated from HTML - Inappropriate header comments - do not match the in-house standard - Code comments inadequate to inform later code understanding/maintenance - Unwell-structured codes, e.g., inappropriate indents	-21	
Wireframe - Wireframe does not match the actual website - Wireframe does not have any labels. - All pages have a wireframe.	-10 -5 Each missing page 3 points	
CSS (deduct up to 3 marks each) - Redundant CSS included or unused selectors included - Inappropriate header comments - do not match the in-house standard - Pages have similar look and flow (up to 10 marks will be deducted) - Code comments are inadequate to inform later code understanding/maintenance - All CSS codes are in CSS files, no CSS code is embedded in HTML	-22	
Website and submission - The directory and file structure are not as specified - Failed to deploy the website on Mercury and add the link to Canvas as a comment when submitting - Failed to demonstrate the work to the tutor or failed to acknowledge third-party code.	-4 -8 -30	
Total Deductions		